

TMUA PRACTICE TEST SERIES

TMUA Mock Test 6

Paper 2 — All Topics

- 20 multiple-choice questions
- Recommended time: 75 minutes
- Paper 2 styled
- Complete verified solutions

SOLUTION BOOK

ThrivingScholars

TMUA Mock Test 6 — Paper 2 (All Topics)

This solution book matches the final 20-question test and provides independently checked, non-calculator worked solutions with the relevant diagrams restored.

Answer key

Q1	D	Q2	B	Q3	A	Q4	C	Q5	A	Q6	D	Q7	B	Q8	D	Q9	C	Q10	E
Q11	E	Q12	D	Q13	E	Q14	D	Q15	D	Q16	B	Q17	D	Q18	E	Q19	C	Q20	C

TMUA PAPER 2

Logic and Implication

Questions 1–6: contrapositives, counterexamples, implication chains and seating constraints.

Q1

Contrapositive

Answer D

Question 1

Which of the following is equivalent to “If P is true, then Q is false”?

A “ P is true or Q is false.”

B “If Q is false, then P is true.”

C “If P is false, then Q is true.”

D “If Q is true, then P is false.”

E “If Q is true, then P is true.”

WORKED SOLUTION

The statement is $P \Rightarrow \neg Q$. Its contrapositive is

$$Q \Rightarrow \neg P.$$

A statement and its contrapositive are logically equivalent. Therefore the correct answer is **D**.

Question 2

The remainder is defined for all real numbers x and y , with $y \neq 0$, by

$$\text{rem}(x, y) = x - y \left\lfloor \frac{x}{y} \right\rfloor,$$

where $\left\lfloor \frac{x}{y} \right\rfloor$ denotes the greatest integer less than or equal to $\frac{x}{y}$.

What is the value of $\text{rem}\left(\frac{3}{8}, -\frac{2}{5}\right)$?

A $-\frac{3}{8}$

B $-\frac{1}{40}$

C 0

D $\frac{3}{8}$

E $\frac{31}{40}$

WORKED SOLUTION

By definition,

$$\text{rem}\left(\frac{3}{8}, -\frac{2}{5}\right) = \frac{3}{8} - \left(-\frac{2}{5}\right) \left\lfloor \frac{\frac{3}{8}}{-\frac{2}{5}} \right\rfloor.$$

Now

$$\frac{\frac{3}{8}}{-\frac{2}{5}} = -\frac{15}{16}, \quad \left\lfloor -\frac{15}{16} \right\rfloor = -1.$$

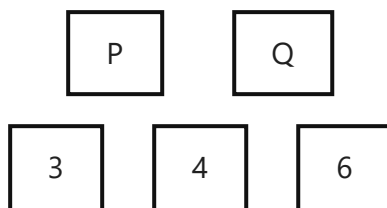
Hence

$$\frac{3}{8} - \left(-\frac{2}{5}\right)(-1) = \frac{3}{8} - \frac{2}{5} = -\frac{1}{40}.$$

The answer is **B**.

Question 3

Five cards are lying on a table as shown.



Each card has a letter on one side and a whole number on the other. Jane says, “If a vowel is on one side of a card, then an even number is on the other side.” Mary proves Jane wrong by turning over one card. Which card does Mary turn over?

A numbered card is identified by the number showing on it; for example, card 4 is the card displaying 4.

A 3

B 4

C 6

D P

E Q

WORKED SOLUTION

Jane’s claim has the form

vowel $\Rightarrow$ even number.

To disprove it, Mary must find a card with a vowel on one side and an odd number on the other. The only visible card that can establish this with one turn is the odd-numbered card 3.

The cards 4 and 6 cannot disprove the implication because an even number may have either a vowel or a consonant on the reverse. The letters **P** and **Q** are consonants, so they also cannot disprove the claim.

The answer is **A**.

Question 4

Alan, Beth, Carlos and Diana are discussing their possible grades. Alan says, "If I get an A, then Beth will get an A." Beth says, "If I get an A, then Carlos will get an A." Carlos says, "If I get an A, then Diana will get an A." All three statements are true, but exactly two students receive an A. Which two receive A grades?

A Alan and Beth

B Beth and Carlos

C Carlos and Diana

D Alan and Diana

E Beth and Diana

WORKED SOLUTION

The statements form the implication chain

$$A \Rightarrow B \Rightarrow C \Rightarrow D.$$

If Alan receives an A, then all four students must receive A grades. If Beth receives an A, then Carlos and Diana must also receive A grades. Therefore neither option can contain Alan or Beth while still giving exactly two A grades.

Carlos and Diana can both receive A grades: Carlos's statement forces Diana's grade, while Alan's and Beth's implications remain true because their premises are false.

The answer is **C**.

Question 5

Consider the statement:

If a whole number n is not prime, then $n - 2$ is not prime.

Which value of n shows that the statement is false?

A 9

B 12

C 13

D 16

E 23

WORKED SOLUTION

An implication is false only when its premise is true and its conclusion is false. We therefore need n to be non-prime while $n - 2$ is prime.

For $n = 9$, the number 9 is not prime, but

$$9 - 2 = 7$$

is prime. Thus $n = 9$ is a counterexample.

The answer is **A**.

Question 6

Abby, Bret, Carl and Dana sit in a row of four seats numbered 1 to 4. Joe says:

“Bret is next to Carl.”

“Abby is between Bret and Carl.”

Both statements are false. Bret is sitting in seat 3. Who is sitting in seat 2?

A Abby

B Bret

C Carl

D Dana

E There is not enough information

WORKED SOLUTION

Bret is in seat 3. Because “Bret is next to Carl” is false, Carl cannot be in seat 2 or seat 4. Hence Carl is in seat 1.

The seats between Carl in seat 1 and Bret in seat 3 consist only of seat 2. Since “Abby is between Bret and Carl” is also false, Abby cannot be in seat 2. Abby must therefore be in seat 4, leaving Dana in seat 2.

The answer is **D**.

Sequences, Probability and Geometry

Questions 7–10: coordinate movement, periodic recurrences, continuous probability and circle geometry.

Q7

Coordinate movement

Answer B

Question 7

Andy the Ant starts at $(-20, 20)$, facing east. He moves 1 unit, turns 90° left, moves 2 units, turns 90° left, moves 3 units, and continues in this way, increasing the distance by 1 unit after each turn.

At what point does Andy make his 2020th left turn?

A $(-1030, -994)$ B $(-1030, -990)$ C $(-1026, -994)$ D $(-1026, -990)$ E $(-1022, -994)$

WORKED SOLUTION

The directions repeat east, north, west, south. Among the first 2020 moves, there are 505 moves in each direction.

The horizontal displacement is

$$(1 + 5 + \cdots + 2017) - (3 + 7 + \cdots + 2019).$$

Each sum has 505 terms, so

$$\frac{505(1 + 2017)}{2} - \frac{505(3 + 2019)}{2} = 505(1009 - 1011) = -1010.$$

Starting from $x = -20$, the final x -coordinate is -1030 .

Similarly, the vertical displacement is

$$(2 + 6 + \cdots + 2018) - (4 + 8 + \cdots + 2020) = -1010,$$

so the final y -coordinate is $20 - 1010 = -990$.

The answer is **B**.

Question 8

Define a sequence recursively by $F_0 = 0$, $F_1 = 1$, and let F_n be the remainder when $F_{n-1} + F_{n-2}$ is divided by 3, for all $n \geq 2$. Thus the sequence begins

$$0, 1, 1, 2, 0, 2, \dots$$

Find

$$F_{2017} + F_{2018} + F_{2019} + F_{2020} + F_{2021} + F_{2022} + F_{2023} + F_{2024}.$$

A 6

B 7

C 8

D 9

E 10

WORKED SOLUTION

Working modulo 3 gives

$$0, 1, 1, 2, 0, 2, 2, 1, 0, 1, 1, 2, \dots$$

The sequence has period 8. Since **2016** is divisible by 8, the terms from F_{2017} to F_{2024} form one complete period:

$$1 + 1 + 2 + 0 + 2 + 2 + 1 + 0 = 9.$$

The answer is **D**.

Question 9

Chloé chooses a real number uniformly at random from $[0, 2017]$. Independently, Laurent chooses a real number uniformly at random from $[0, 4034]$.

What is the probability that Laurent's number is greater than Chloé's number?

A $\frac{1}{2}$

B $\frac{2}{3}$

C $\frac{3}{4}$

D $\frac{5}{6}$

E $\frac{7}{8}$

WORKED SOLUTION

Scale both intervals by 2017. Let X be uniform on $[0, 1]$ and Y uniform on $[0, 2]$. We require $Y > X$.

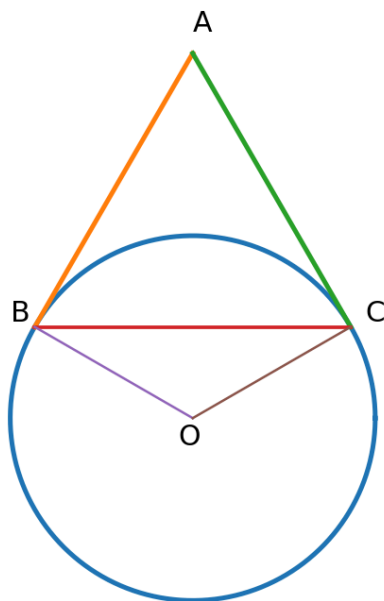
If $Y \in [1, 2]$, which occurs with probability $\frac{1}{2}$, then $Y > X$ automatically. If $Y \in [0, 1]$, the two values are identically and independently distributed on $[0, 1]$, so $Y > X$ with probability $\frac{1}{2}$.

$$P(Y > X) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}.$$

The answer is C.

Question 10

Sides $\overline{AB}$ and $\overline{AC}$ of equilateral triangle ABC are tangent to a circle at B and C , respectively.



What fraction of the area of $\triangle ABC$ lies outside the circle?

A $\frac{4\sqrt{3}\pi}{27} - \frac{1}{3}$

B $\frac{\sqrt{3}}{2} - \frac{\pi}{8}$

C $\frac{1}{2}$

D $\sqrt{3} - \frac{2\sqrt{3}\pi}{9}$

E $\frac{4}{3} - \frac{4\sqrt{3}\pi}{27}$

WORKED SOLUTION

Let the circle have radius r and centre O . Since AB and AC are tangents, $OB \perp AB$ and $OC \perp AC$. The quadrilateral $ABOC$ therefore gives

$$\angle BOC = 120^\circ.$$

Hence the chord

$$BC = 2r \sin 60^\circ = \sqrt{3}r.$$

Since ABC is equilateral, its area is

$$[ABC] = \frac{\sqrt{3}}{4}(\sqrt{3}r)^2 = \frac{3\sqrt{3}}{4}r^2.$$

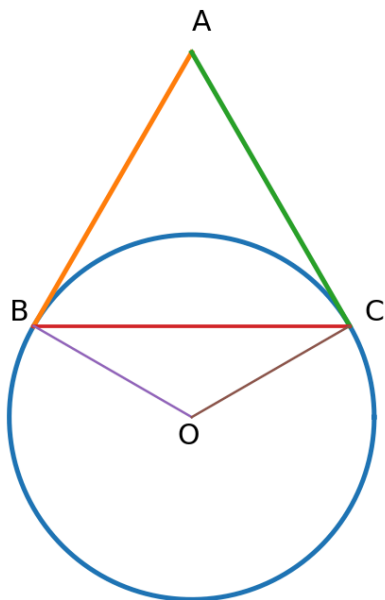
The part of the circle inside the triangle is the minor segment cut off by chord BC :

$$\frac{120^\circ}{360^\circ}\pi r^2 - \frac{1}{2}r^2 \sin 120^\circ = \frac{\pi r^2}{3} - \frac{\sqrt{3}r^2}{4}.$$

Thus the fraction of the triangle outside the circle is

$$\frac{\frac{3\sqrt{3}}{4}r^2 - \left(\frac{\pi r^2}{3} - \frac{\sqrt{3}}{4}r^2\right)}{\frac{3\sqrt{3}}{4}r^2} = \frac{4}{3} - \frac{4\sqrt{3}\pi}{27}.$$

The answer is **E**.



The radii to the tangent points are perpendicular to the two equal sides.

Algebra and Data

Q11

Inequality reasoning

Answer E

Question 11Real numbers x, y, z satisfy

$$0 < x < 1, \quad -1 < y < 0, \quad 1 < z < 2.$$

Which expression is necessarily positive?

A $y + x^2$

B $y + xz$

C $y + y^2$

D $y + 2y^2$

E $y + z$

WORKED SOLUTIONBecause $-1 < y < 0$ and $1 < z < 2$, we have

$$y + z > -1 + 1 = 0.$$

The inequality is strict, so $y + z$ is always positive.The other options can be non-positive for suitable allowed values. Therefore the answer is **E**.

Question 12

Suppose that non-zero real numbers x and y satisfy

$$\frac{3x + y}{x - 3y} = -2.$$

Find

$$\frac{x + 3y}{3x - y}.$$

A -3

B -1

C 1

D 2

E 3

WORKED SOLUTION

Rearranging the given equation,

$$3x + y = -2(x - 3y) = -2x + 6y.$$

Therefore

$$5x = 5y \implies x = y.$$

Substituting into the required ratio gives

$$\frac{x + 3y}{3x - y} = \frac{4x}{2x} = 2.$$

The answer is **D**.

Question 13

The lines

$$ax - 2y = c \quad \text{and} \quad 2x + by = -c$$

are perpendicular and intersect at $(1, -5)$. Find c .

A -13

B -8

C 2

D 8

E 13

WORKED SOLUTION

The first line has gradient $a/2$, and the second has gradient $-2/b$. Perpendicularity gives

$$\frac{a}{2} \left(-\frac{2}{b} \right) = -1 \implies a = b.$$

Since $(1, -5)$ lies on both lines,

$$a + 10 = c, \quad 2 - 5a = -c.$$

Substituting $c = a + 10$ into the second equation,

$$2 - 5a = -a - 10 \implies 12 = 4a \implies a = 3.$$

Hence $c = a + 10 = 13$. The answer is **E**.

Question 14

At Typico High School, 60% of students like dancing and the remaining 40% dislike it. Of those who like dancing, 80% say that they like it. Of those who dislike dancing, 90% say that they dislike it.

What fraction of the students who say they dislike dancing actually like it?

A 10%

B 12%

C 20%

D 25%

E $33\frac{1}{3}\%$

WORKED SOLUTION

The proportion who genuinely like dancing but say they dislike it is

$$0.60 \times 0.20 = 0.12.$$

The proportion who genuinely dislike dancing and say they dislike it is

$$0.40 \times 0.90 = 0.36.$$

Therefore 0.48 of all students say they dislike dancing, and the required conditional proportion is

$$\frac{0.12}{0.12 + 0.36} = \frac{1}{4} = 25\%.$$

The answer is **D**.

Question 15

The mean, median and mode of the seven values

$$60, 100, x, 40, 50, 200, 90$$

are all equal to x . Find x .

A 50

B 60

C 75

D 90

E 100

WORKED SOLUTION

If the mean is x , then

$$\frac{60 + 100 + x + 40 + 50 + 200 + 90}{7} = x.$$

Thus

$$540 + x = 7x \implies x = 90.$$

The resulting ordered data are

$$40, 50, 60, 90, 90, 100, 200,$$

so the median and mode are also 90. The answer is **D**.

Geometry and Polynomial Structure

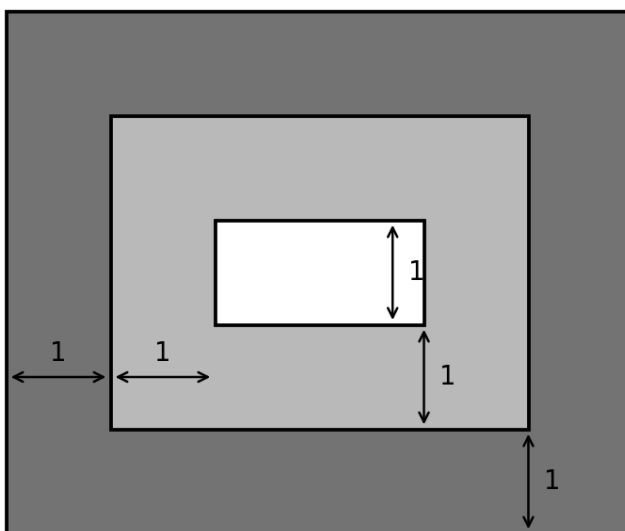
Questions 16–20: nested areas, shaded regions, integer averages, hyperbolas and polynomial divisibility.

Q16

Areas in arithmetic progression

Answer B**Question 16**

A rug is made with three differently coloured regions as shown. The areas of the three regions form an arithmetic progression. The inner rectangle is 1 foot wide, and each surrounding region is 1 foot wide on all four sides.



What is the length, in feet, of the inner rectangle?

A 1**B** 2**C** 4**D** 6**E** 8

WORKED SOLUTION

Let the inner rectangle have length x . Its area is x .

The middle rectangle has dimensions $(x + 2)$ by 3, so the area of the middle coloured region is

$$3(x + 2) - x = 2x + 6.$$

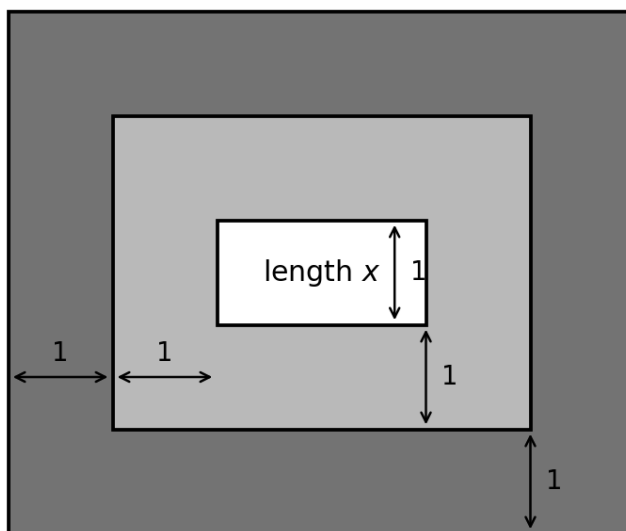
The outer rectangle has dimensions $(x + 4)$ by 5, so the area of the outer coloured region is

$$5(x + 4) - 3(x + 2) = 2x + 14.$$

These three regional areas form an arithmetic progression, so

$$2(2x + 6) = x + (2x + 14).$$

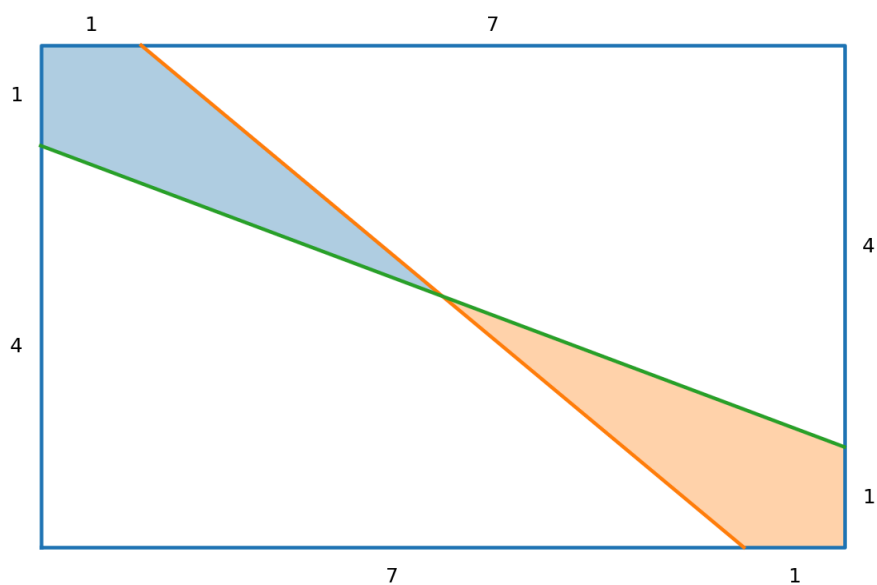
Hence $4x + 12 = 3x + 14$, giving $x = 2$. The answer is **B**.



The successive rectangles have dimensions $x \times 1$, $(x + 2) \times 3$, and $(x + 4) \times 5$.

Question 17

Find the area of the shaded region.



A $4\frac{3}{5}$

B 5

C $5\frac{1}{4}$

D $6\frac{1}{2}$

E 8

WORKED SOLUTION

Take the rectangle to have width 8 and height 5. The two lines meet at the centre $(4, \frac{5}{2})$: one joins $(1, 5)$ to $(7, 0)$, and the other joins $(0, 4)$ to $(8, 1)$.

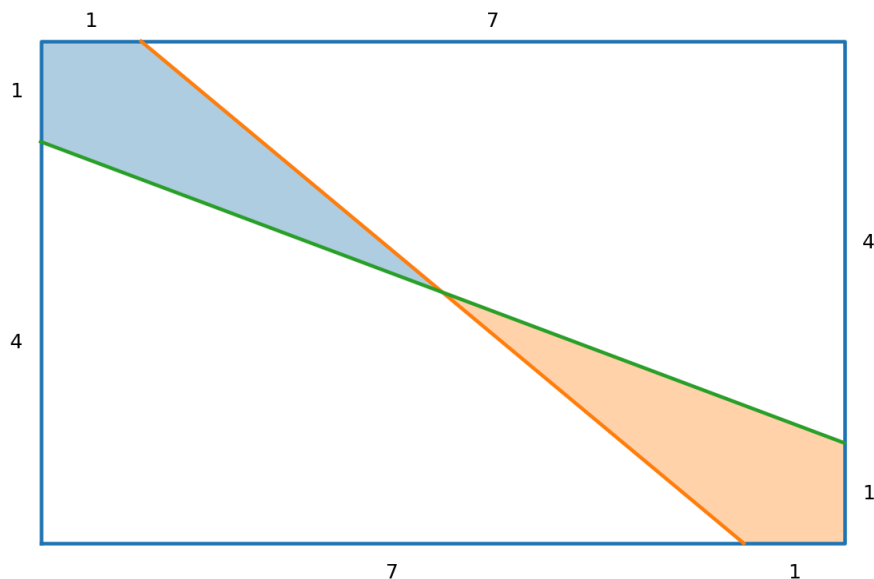
Split each shaded corner region into two triangles. At the upper-left corner, one triangle has base 1 and perpendicular height 4, while the other has base 1 and height $\frac{5}{2}$. Their combined area is

$$\frac{1}{2}(1)(4) + \frac{1}{2}(1)\left(\frac{5}{2}\right) = 2 + \frac{5}{4}.$$

By half-turn symmetry, the lower-right shaded region has the same area. Therefore the total shaded area is

$$2\left(2 + \frac{5}{4}\right) = \frac{13}{2} = 6\frac{1}{2}.$$

The answer is **D**.



The lines intersect at $(4, \frac{5}{2})$, allowing each shaded region to be split into two triangles.

Question 18

Isabella takes seven mathematics tests and receives seven distinct integer scores, each between 91 and 100 inclusive. After each test, the average of all her scores so far is an integer. Her seventh score is 95.

What is her sixth score?

A 92

B 94

C 96

D 98

E 100

WORKED SOLUTION

Let S_k be the sum after k tests. Because the average after seven tests is an integer, S_7 is divisible by 7.

Seven distinct scores from 91 to 100 have a total between

$$91 + 92 + \cdots + 97 = 658$$

and

$$94 + 95 + \cdots + 100 = 679.$$

The multiples of 7 in this interval are **658, 665, 672, 679**. Since the seventh score is 95,

$$S_6 = S_7 - 95 \in \{563, 570, 577, 584\}.$$

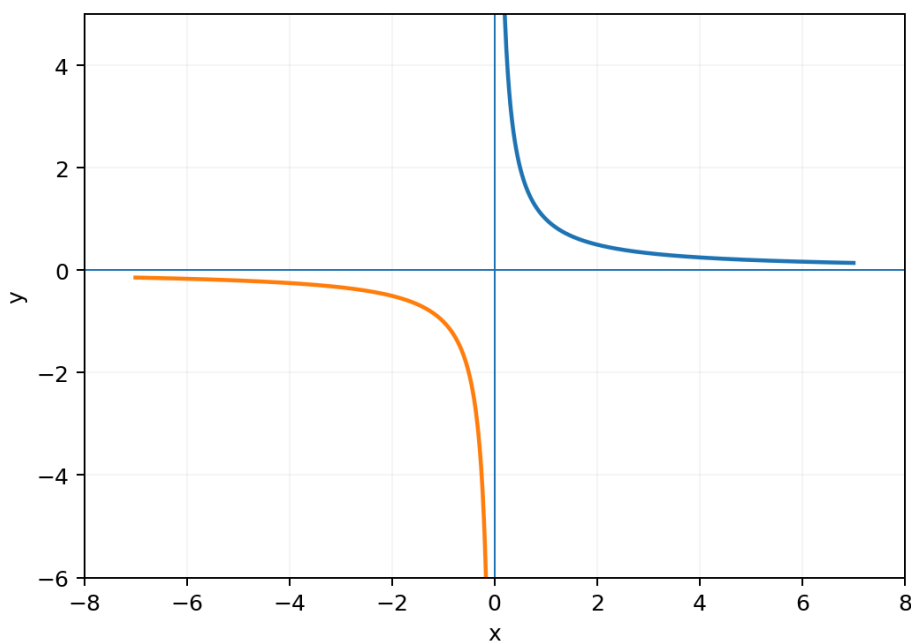
The average after six tests is an integer, so S_6 must be divisible by 6. Only 570 works.

Now $S_5 = 570 - s_6$ is divisible by 5. Hence s_6 must be divisible by 5. The only possible scores are 95 and 100, and 95 has already been used as the seventh score. Therefore $s_6 = 100$.

The answer is **E**.

Question 19

The vertices of an equilateral triangle lie on the hyperbola $xy = 1$, and a vertex of the hyperbola is the centroid of the triangle.



What is the square of the area of the triangle?

A 48

B 60

C 108

D 120

E 169

WORKED SOLUTION

Without loss of generality, let the hyperbola vertex that is the centroid be

$$I = (-1, -1).$$

For an equilateral triangle, the centroid is also the circumcentre. Let the circumcircle have radius R . A point (x, y) on both the circle and the hyperbola satisfies

$$xy = 1$$

and

$$(x + 1)^2 + (y + 1)^2 = R^2.$$

Writing $s = x + y$, the circle equation becomes

$$s^2 + 2s = R^2.$$

The circle must meet the hyperbola in exactly three distinct points. One of the two possible values of s must therefore produce a repeated intersection. This occurs at the opposite hyperbola vertex $(1, 1)$, where $s = 2$. Hence

$$R^2 = 2^2 + 2(2) = 8, \quad R = 2\sqrt{2}.$$

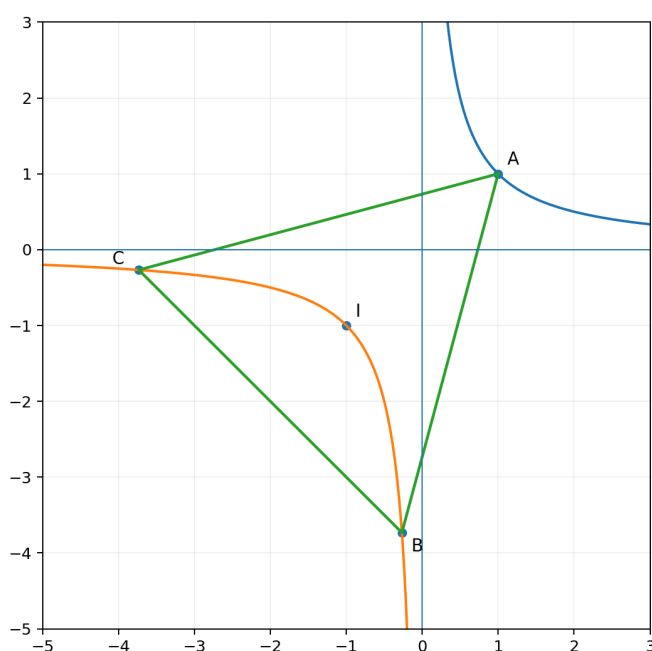
The side length a of an equilateral triangle is $a = R\sqrt{3} = 2\sqrt{6}$. Its area is

$$\frac{a^2\sqrt{3}}{4} = \frac{24\sqrt{3}}{4} = 6\sqrt{3}.$$

The square of the area is

$$(6\sqrt{3})^2 = 108.$$

The answer is **C**.



The circumcentre is the hyperbola vertex $(-1, -1)$; the opposite vertex $(1, 1)$ fixes the circumradius.

Question 20

For certain real numbers a, b, c , the polynomial

$$g(x) = x^3 + ax^2 + x + 10$$

has three distinct roots, and every root of g is also a root of

$$f(x) = x^4 + x^3 + bx^2 + 100x + c.$$

Find $f(1)$.

A -9009

B -8008

C -7007

D -6006

E -5005

WORKED SOLUTION

Since g is a monic cubic with three distinct roots and every root is also a root of the monic quartic f , the polynomial g divides f . Therefore, for some real k ,

$$f(x) = (x + k)g(x).$$

Expanding,

$$(x + k)(x^3 + ax^2 + x + 10) = x^4 + (a + k)x^3 + (1 + ak)x^2 + (10 + k)x + 10k.$$

Comparing the coefficient of x with $f(x)$ gives

$$10 + k = 100 \implies k = 90.$$

Comparing the coefficient of x^3 ,

$$a + k = 1 \implies a = -89.$$

Now

$$g(1) = 1 - 89 + 1 + 10 = -77,$$

and hence

$$f(1) = (1 + 90)g(1) = 91(-77) = -7007.$$

The answer is **C**.

